MIL-B-48008 (AR)
7 July 1977
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

BATTERY, THERMAL, XM97

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of battery entitled Battery, Thermal, XM97 which is used with the Sprint warhead.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-W-6858	- Welding, Resistance, Nickel Alloys.
MIL-W-8611	- Welding, Metal Arc and Gas, Steels, and Corrosion and Heat Resistant Alloys, Process for.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202	- Test Method for Electronic and Electrical Component Parts.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure for.

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

FSC 1135

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DRAWINGS

PICATINNY ARSENAL

9274396	- Box, Fiberboard Packing for Battery Thermal: XM-97.
9274509	- Battery Thermal: XM-97.
EL-9274467	- Equipment List.
EL-9274468	- Equipment List.
EL-9274469	- Equipment List.
EL-9274509	- Equipment List.

PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1	- Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The battery shall comply with all requirements specified on the drawing (dwg) 9274509, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A serial number shall be assigned to the item. Serial numbers required on these items shall be obtained by means of a letter addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial numbers required.

3.5 Coded serialization requirements (see 6.1.2). The contractor shall supply feedback data on punch cards for any item for which coding is required in section 3. The feedback data shall be in accordance with QAI-PA-1.

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3.6 Insulation resistance. The insulation resistance between each isolated circuit and all others and case shall be 50 megohms minimum when tested with 500±50 volts dc between the points specified in Table I and as specified in 4.4.1.

Table I

Note: Shorting strips must be between both positive terminals to both negative terminals prior to hypot. Short strips must also be applied from Scom to S_1 and S_2 respectively. This is accomplished automatically by AIE.

a. Positive (+) to Match (Scom)
b. Positive (+) to Monitor (M_1)
c. Positive (+) to Case
d. Monitor (M_1) to Match (Scom)
e. Monitor (M_1) to Case
f. Match (Scom) to Case

3.7 Circuit resistance. The circuit resistance of each circuit shall be within the values listed in Table II and Table III when tested as specified in 4.4.2.

3.7.1 Prior to activation.

Table II

Terminals	Resistance (Ohms)
a. S_1 to Scom	5.3 ± 0.3
b. S_2 to Scom	5.3 ± 0.3
c. M_1 to M_2	Less than 1.0
d. + to -	1.7 ± 0.2
e. + to -	1.7 ± 0.2

3.7.2 After activation.

Table III

Terminals	Resistance (Ohms)
M_1 to M_2	Greater than 100,000

3.8 Activation.

3.8.1 All-fire current. Each match shall activate when a current of 500 ± 50 milliamperes dc is passed through the activation circuit for a maximum of 50 milliseconds. After the activation the battery shall comply with 3.7.2

3.8.2 No-fire current. Each match shall not activate or be damaged (see 6.5) when current of 250 + 0, -5 milliamperes dc is passed through the activation circuit for a minimum of 5 seconds when tested as specified in 4.4.4. After completion of this test, the battery shall comply with resistance requirements of M_1 to M_2 of 3.7.1.

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3.9 Leak test. The battery shall exhibit no evidence of leakage when tested as specified in 4.4.4.

3.10 Electrical performance. The battery at a temperature of 120±5°F shall comply with the voltage requirements of Figure 2 when loaded as specified in Figure 2. The tolerance on the load duration is +0%, -5%. When a battery is tested for these requirements, it will be tested as specified in 4.4.5. Record required data using the code numbers specified in Figure 2 with the last digit to be the test condition digit specified in Table IV. After this test the battery shall comply with the requirement of 3.7.2.

Table IV

Test Condition	Code No.
Bench at 120°F	0
Lateral Acceleration at 120°F (Z-Z axis)	1
Longitudinal Acceleration at 120°F (X-X axis)	2

3.11 High temperature. The battery shall withstand a temperature of 165±5°F for twelve (12) hours minimum without evidence of physical damage (see 6.5) when tested as specified in 4.4.6. After completion of this test the battery shall comply with the requirements of 3.6 and 3.7.1.

3.12 Low temperature. The battery shall withstand a temperature of -65±5°F for twelve (12) hours minimum without evidence of physical damage (see 6.5) when tested as specified in 4.4.7. After completion of this test the battery shall comply with the requirements of 3.6 and 3.7.1.

3.13 Temperature shock. The battery shall withstand a temperature shock from 165±5°F to -65±5°F to 165±5°F without evidence of physical damage (see 6.5). Testing shall consist of three (3) cycles as outlined in 4.4.8. After completion of this test the battery shall comply with the requirements of 3.6, 3.7.1 and 3.9.

3.14 Transportation vibration. The battery shall withstand the vibration spectrum in Table V at room temperature for (1) minute along each of its mutually perpendicular axes (see Figure 1) without evidence of physical damage (see 6.5). The vibration test shall be in accordance with 4.4.9. At the completion of this test the battery shall comply with requirements of 3.9 and 3.7.1.

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Table V

Frequency (Hz)	Amplitude
5 - 10	1.0 in d.a.
10 - 100	5.0g
100 - 500	1.0g

3.15 Mechanical shock. The battery shall withstand shock impulses of 65±5 g's three (3) times in each direction in each of its two mutually perpendicualr axes (see Figure 1) for a total of 12 shocks without evidence of physical damage (see 6.5). A one-half sine pulse shall be used with a duration of 11±2 milliseconds. Testing shall be in accordance with 4.4.10. After completion of this test the battery shall comply with the requirements of 3.6, 3.7.1 and 3.9.

3.16 Acceleration.

3.16.1 Acceleration operating. The battery shall be activated and electrical performance tested in accordance with 3.10 and 4.4.5. Testing shall be in accordance with 4.4.12.

3.16.2 Acceleration lateral. The battery shall be activated and electrical performance tested in accordance with 3.10 and 4.4.5. Simultaneously the battery shall be subjected to a lateral (Z-Z) acceleration of 78 ± 2 g's for seven (7) seconds minimum, while at a temperature of 120 ± 5°F. Testing shall be in accordance with 4.4.11. After completion of this test the battery shall comply with the requirements of 3.7.2.

3.16.3 Acceleration on longitudinal. The battery shall be activated and electrical performance tested in accordance with 3.10 and 4.4.5. Simultaneously the battery shall be subjected to a longitudinal (X-X) acceleration force of 170±2 g's for 2.5 seconds minimum, while at a temperature of 120±5°F. Testing shall be in accordance with 4.4.11. After completion of this test the battery shall comply with the requirements of 3.7.2.

3.17 Welding. All welding shall comply with the requirements specified in MIL-W-6858 or MIL-W-8611 as applicable.

3.18 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.19 X-ray. The battery shall not show a positive or negative lead extending into the area between the header potting compound and the case when X-rayed as specified in 4.4.13.

3.20 Workmanship.

3.20.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.20.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.20.3 Burrs. Parts shall be free from burrs.

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3.20.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.20.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 9 complete assemblies (see dwg 9274509) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table VI. Tests shall be conducted in order specified in Table VI.

Table VI

First article inspection and tests

Inspection or Test	Unit No.'s To Be Tested	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)	all		X
2. X-ray (see 3.19)			X
3. Insulation resistance (see 3.6)	all		X
4. Circuit resistance (see 3.7)	all		X
5. Mechanical shock (see 3.15)	all		X
6. High temperature (see 3.11)	all		X
7. Low temperature (see 3.12)	all		X
8. Temperature shock (see 3.13)	all		X
9. Transportation vibration (see 3.14)	all		X
10. No fire current (see 3.8.2)	all		X
11. Leak test (see 3.9)	all		X
12. Acceleration operating (see 3.16.1)	1	X	
13. Acceleration longitudinal (see 3.16.3)	4-6	X	

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Inspection or Test	Unit No.'s To Be Tested	Dest	Non-Dest
14. Acceleration operating (see 3.16.1)	2	X	
15. Acceleration lateral (see 3.16.2)	7-9	X	
16. Acceleration operating (see 3.16.1)	3	X	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor defects/characteristics table (see Table VII and Table VIII), and inspected for all defects/characteristics in the applicable classification. Any defects/characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of defects/characteristics in each classification need not be followed as long as all defects/characteristics inspections are performed.

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Table VII
Major Defects/Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table VIII
Minor Defects/Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of defects/characteristics (see 3.2). Critical, major and minor defects/characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Cell stack assembly, pulse (see dwg 9274467).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
*101.	Incorrect weight of depolarizer pads	Gage
*102.	Incorrect weight of electrolyte pads	Gage
103.	Improper assembly of cell components	Visual
104.	Defective or loose grommet	Visual
105.	Improper orientation of heat pad inert faces	Visual
*106.	Verification of calorific value of heat pad pairs	Gage
107.	Improper orientation and position of insulator assembly and leads	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.20)	Visual

*To be accomplished by verification of records, for each lot of pads, prior to assembly into batteries.

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4.3.2.1.2 Cell stack assembly, power (see dwg 9274468).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
*101.	Incorrect weight of depolarizer pads	Gage
*102.	Incorrect weight of electrolyte pads	Gage
103.	Improper assembly of cell components	Visual
104.	Defective or loose grommet	Visual
105.	Improper orientation of heat pad inert faces	Visual
*106.	Verification of calorific value of heat pad pairs	Gage
107.	Improper orientation and position of insulator assembly and leads	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.20)	Visual

4.3.2.1.3 Final header assembly (see dwg 9274469).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Insulation resistance not in accordance with drawing requirements	Gage
102.	Circuit resistance not in accordance with drawing requirements	Gage
Minor:		
201.	Evidence of poor workmanship (see 3.20)	Visual

*To be accomplished by verification of records, for each lot of pads, prior to assembly into batteries.

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4.3.2.1.4 Battery, completed (see dwg 9274509).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Location of header pins	Gage
102.	Location of mounting holes	Gage
103.	Diameter of mounting holes	Gage
104.	Alignment of Scom pin with closest mounting hole	Gage
105.	Height of mounting bracket off body	Gage
106.	Outside diameter of body	Gage
107.	Height of pins	Gage
108.	Straightness of bracket mounting surfaces to each other	Gage
109.	Distance of bracket mounting surface to pin base surface	Gage
110.	Location of flat on inside bolting surface	Gage
111.	Bent pins	Visual
112.	Diameter of header pins	Gage
113.	All pins shorted together	Visual
114.	Temp indicating paint and lable missing	Visual
*115.	Evidence of cracks in glass seals of header	Visual
116.	Serialization improper or missing (see 3.5)	Visual
117.	Objective evidence relative to material missing or improper (see 3.1)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.20)	Visual

4.3.2.1.5 Preparation for delivery (see 5).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Packaging, packing and marking not in accordance with drawing 9274396	Visual
Minor:	None defined.	

*This inspection shall be performed using an optical instrument of 20±5 power magnification.

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4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test classification
a. X-ray	Major
b. Insulation resistance	Major
c. Circuit resistance	Major
d. Leak test	Major
e. No-fire current	Major
f. acceleration operating	Major
g. acceleration longitudinal	Major
h. acceleration lateral	Major

Of these tests, e, f, g, and h are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 X-ray, insulation resistance, circuit resistance, and leak test. All batteries shall be tested in accordance with 4.4.13, 4.4.1, 4.4.2, and 4.4.4. If any item fails to comply with 3.19, 3.6, 3.7, and 3.9, the item shall be rejected and removed from the lot.

4.3.3.2 No-fire current, acceleration operating, acceleration longitudinal, and acceleration lateral. A sample shall be selected at random in accordance with Table IX. The sample shall be tested in accordance with 4.4.3, 4.4.5, and 4.4.12, 4.4.5 and 4.4.11. If any item fails to comply with 3.19, 3.6, 3.7 and 3.9, the lot shall be rejected.

Table IX

Production Lot Acceptance

No. of Acceptable Units Required to be Delivered	No. of Sample Units to be Destructively Tested (Sample Size)	Inspection Total
1 - 8	8	9 - 16
9	9	18
10 - 18	10	20 - 28
19 - 28	11	30 - 39
29 - 57	12	41 - 69
58 - 429	13	71 - 442
430 and up	14	444 and up

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4.3.4 Acceptance inspection equipment.

4.3.4.1 Major defects/characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-9274509, 9274467, 9274468 and 9274469). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor defects/characteristics and tests. Acceptance inspection equipment for minor defects/characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Insulation resistance. The insulation resistance shall be measured in accordance with MIL-STD-202, Method 302, Test Condition B for 10 seconds minimum or until the meter stabilizes. The current shall be limited to 5 milliamperes maximum. The battery voltage terminals shall be electrically connected together and treated as one electrically isolated circuit. The battery activation terminals shall be treated in the same manner.

4.4.2 Circuit resistance. The dc resistance of each circuit shall be measured in accordance with MIL-STD-202, Method 303. The test current shall not exceed 10 ma and open circuit test voltage shall not exceed 15 volts dc.

4.4.3 No-fire current. The no-fire current requirements of each match shall be tested by using a power supply, approved by the procuring agency, to pass a current of 250 (+0, -5) milliamperes, through the battery activation circuit for a minimum of 5 seconds when determining compliance with 3.8.2. At completion of test, the battery shall be tested in accordance with 4.4.2 to determine if it has activated.

4.4.4 Leak test. The battery shall be placed in a pressure chamber and subjected to a minimum of 90 (psi) of helium gas for a minimum time period of 30 minutes. The battery shall then be removed from the chamber and immediately be placed in a hot water bath at a temperature of 170 + 5°F. The battery shall rest in this water bath for a minimum time period of five minutes. If the battery shows any leaks during

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this five (5) minute period as evidenced by the evolution of a stream of gas bubbles in the hot water bath, the battery shall be rejected. The battery shall not stay submerged for more than 10 minutes during any one testing operation, when determining compliance with 3.9.

4.4.5 Electrical performance. The battery shall be placed into an approved temperature chamber and the temperature raised to 120 + 5°F. After the temperature in the chamber has been stabilized (see 6.4), the battery shall be exposed to this temperature for 4 hours minimum. Within 8 minutes after removing the battery from the temperature chamber, the battery shall be activated by applying simultaneously the all-fire current specified in 3.8.1 to each match circuit. The battery shall be loaded according to the time and load sequence specified in Figure 2. Voltage data shall be continuously recorded to an accuracy of ± 0.25 volts. Current data shall also be continuously recorded to an accuracy of ± 0.15 amperes.

4.4.6 High temperature. The battery shall be placed into an approved temperature chamber and the temperature raised to 165 + 5°F. A relative humidity of 15% maximum shall be maintained in the test chamber. After the temperature in the chamber has stabilized (see 6.4), the battery shall be exposed to this temperature for 4 hours minimum and then returned to ambient room temperature without forced cooling. The battery shall then be examined for any physical damage (see 6.5) and tested in accordance with 4.4.1 and 4.4.2.

4.4.7 Low temperature. The battery shall be placed into an approved temperature chamber and the temperature lowered to -65 + 5°F. After the temperature in the chamber has stabilized (see 6.4), the battery shall be exposed to this temperature for 12 hours minimum and then returned to ambient room temperature without forced heating. The battery shall then be examined for any physical damage (see 6.5) and tested in accordance with 4.4.1 and 4.4.2.

4.4.8 Temperature shock. The battery shall be subjected to three temperature change cycles. Each cycle shall consist of the following: Four hours minimum in the chamber described in 4.4.6. Remove the battery and within one minute transfer the battery to the chamber described in 4.4.7. After four hours minimum remove the battery from the chamber and within one minute transfer the battery to the chamber described in 4.4.6. After four hours minimum, remove the battery from the chamber and examine the battery for physical damage (see 6.5), and test in accordance with 4.4.1, 4.4.2 and 4.4.4.

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4.4.9 Transportation vibration. The battery shall be mounted on a fixture capable of being oriented in each axis (see Figure 1). The fixture with the mounted battery shall be secured on a vibration table, approved by the procuring agency, and subjected to the vibration sequence described in 3.14. After completion of the test, the battery shall be examined for any physical damage (see 6.5) and tested in accordance with 4.4.2 and 4.4.4.

4.4.10 Mechanical shock. The battery shall be mounted on a fixture capable of being oriented in each axis. The fixture, with the mounted battery, shall be secured on the shock table, approved by the procuring agency, and subjected to the shock pulse as specified in 3.15. After shock, the battery shall be examined for physical damage (see 6.5) and tested in accordance with 4.4.1, 4.4.2 and 4.4.4.

4.4.11 Acceleration. The battery, after being stabilized at 120 + 5°F, shall be removed from the chamber, mounted on an approved centrifuge in either the XX or ZZ axes. The battery shall then be activated and subjected to the "G" level and time specified in 3.16. While being subjected to this test, the battery shall be activated and tested in accordance with 4.4.5. Time shall be counted from this point.

4.4.12 Acceleration. The battery shall be mounted on the centrifuge in the -Z to +Z axes. The centrifuge shall remain in the static condition. The battery shall be activated and electrical performance tested in accordance with 3.10 and 4.4.5.

4.4.13 X-ray. The battery shall be X-rayed as viewed through the ZZ axis (+ to -), (see Figure 1).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The battery shall be packaged in accordance with dwg 9274396 using the specified moisture barrier.

5.1.2 Level C. The battery shall be packaged in accordance with dwg 9274396 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The battery shall be packed in accordance with dwg 9274396.

5.3 Marking. Marking shall be as specified in accordance with dwg 9274396.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.2.1). Data specified in 3.5 and 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Physical damage. Defects such as cracks, splitting, distortion, or loosening of parts that are detectable by visual observation are evidence of physical damage.

6.6 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2234 dated 11 October 1972 and SCN 1, 2 and 3 dated 9 October 1974.

Custodian:
Army - AR

Preparing activity:
Army - AR

Project No. 1135-A221

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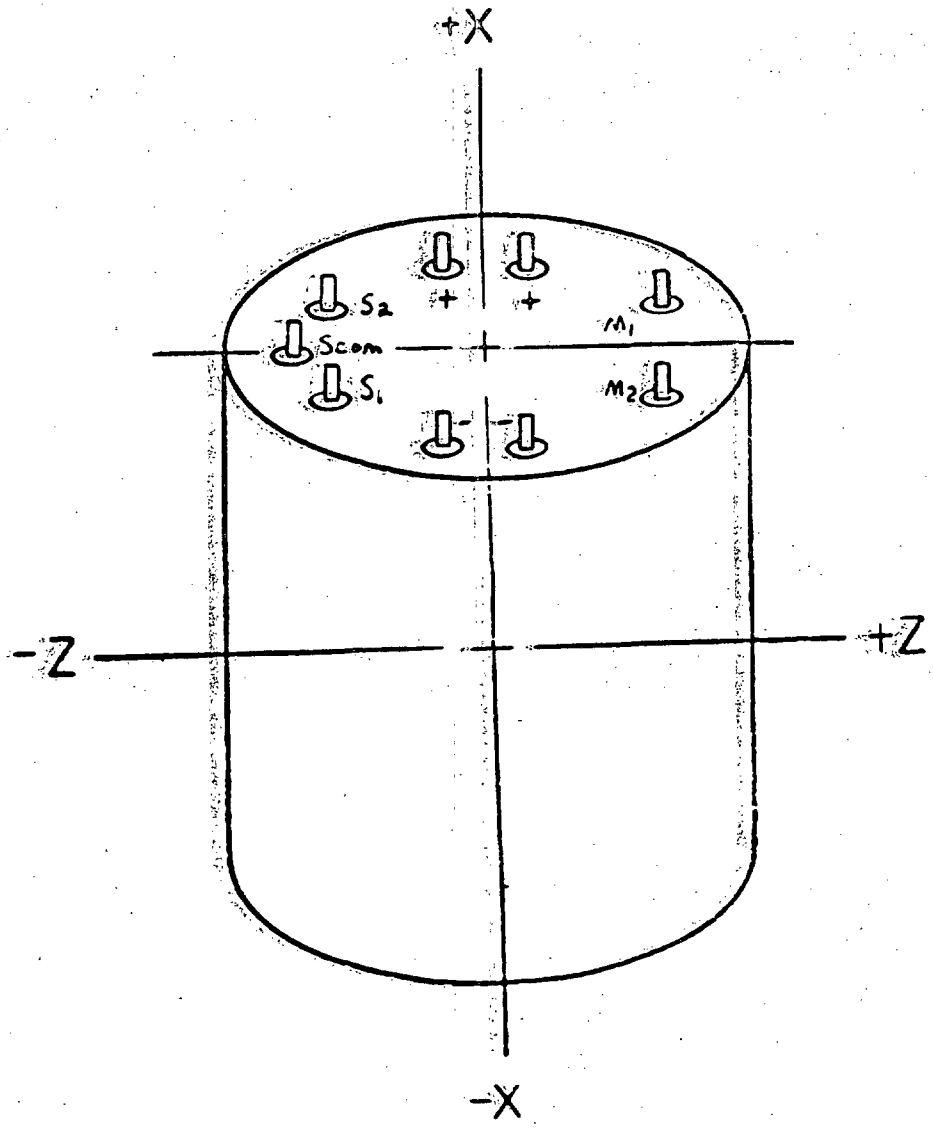


Figure 1. Principal Test Axes

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Event	Nominal Time (Sec.)	Load Resistance (Ohm)	Voltage Requirement (Volts)	Load Duration (Seconds)	Test Code No. (See 3.5 and Table IV) (Begin)	Test Code No. (See 3.5 and Table IV) (End)
1	0 - 800 ms	Open circuit	0 - 37	800 ms	10010-	10011-
2	800 - 900 ms	3.0	23.0 - 36	100 ms	10020-	10021-
3	900 ms - 1.4 Sec.	Open circuit	30.0 - 36	1.5 Sec.	10030-	10031-
4	1.4 - 1.6	See Fig. 3	26.0 - 33.5	200 ms	10040-	10041-
5	1.6 - 1.8	See Fig. 3	29.0 - 33.5	200 ms	10050-	10051-
6	1.8 - 1.85	18	29.0 - 35.0	50 ms	10060-	10061-
7	1.85 - 3.0	70	30.0 - 36.0	1.15 Sec.	10070-	10071-
8	3.0 - 3.2	See Fig. 3	26.0 - 33.5	200 ms	10080-	10081-
9	3.2 - 3.4	See Fig. 3	29.0 - 33.5	200 ms	10090-	10091-
10	3.4 - 3.45	18	29.0 - 35.0	50 ms	10100-	10101-
11	3.45 - 4.0	70	30.0 - 36.0	55 ms	10110-	10111-
12	4.0 - 4.35	8	29.0 - 36.0	350 ms	10120-	10121-
13	4.35 - 59.0	70	30 - 36	54.65 Sec.	10130-	10131-
14	59.0 - 59.2	See Fig. 3	26 - 34	200 ms	10140-	10141-
15	59.2 - 59.4	See Fig. 3	29 - 34	200 ms	10150-	10151-

Figure 2. Time Load Sequence (sheet 1 of 2)

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Event	Nominal Time (Sec.)	Load Resistance (Ohm)	Voltage Requirement (Volts)	Load Duration (Seconds)	Test Code No. (See 3.5 and Table IV) (Begin)	Test Code No. (See 3.5 and Table IV) (End)
16	59.4 - 59.45	18	29 - 35	50 ms	10160-	10161-
17	59.45 - 60.0	70	30 - 36	55 ms	10170-	10171-
18	60.0 - 60.36	8	29 - 35	350 ms	10180-	10181-

NOTE 1 Rise time is 800 ms maximum at which time the battery must be capable of sustaining the load of Event 2.

NOTE 2 Required battery life is 60.36 Sec. However, end of life will be recorded at the point at which the battery (8 ohm) drops to or below 28.5 volts.

Figure 2 (Cont'd). Time Load Sequence (sheet 2 of 2)

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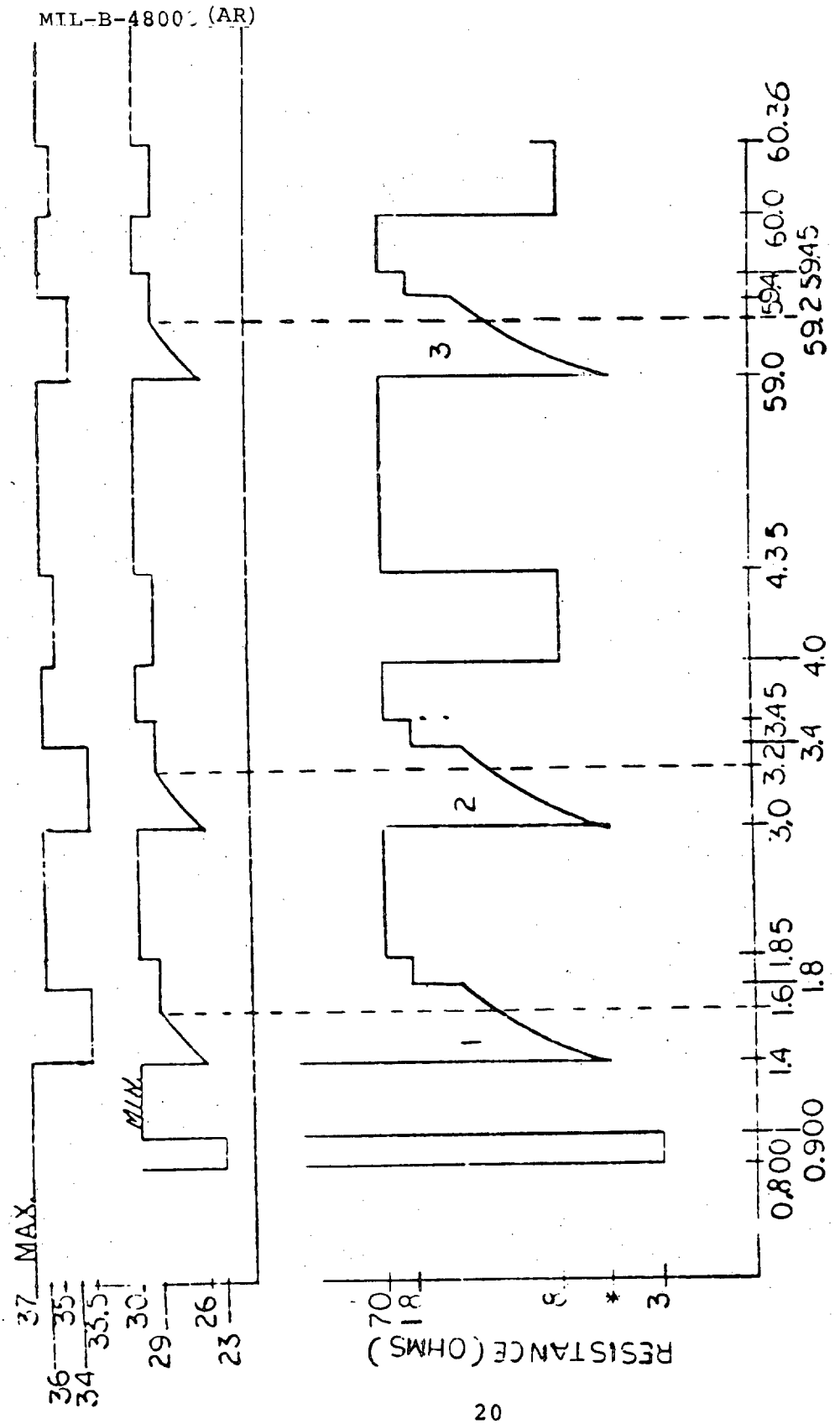
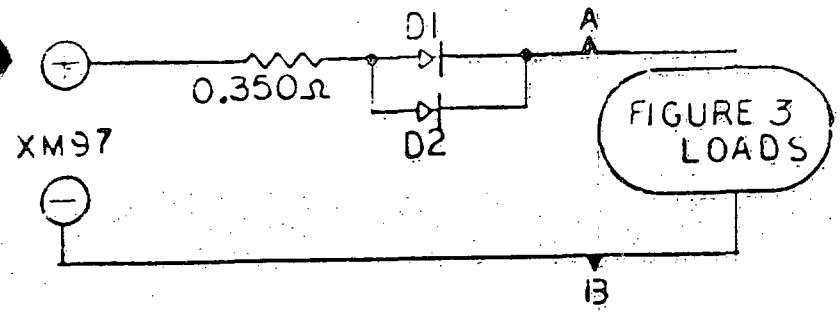


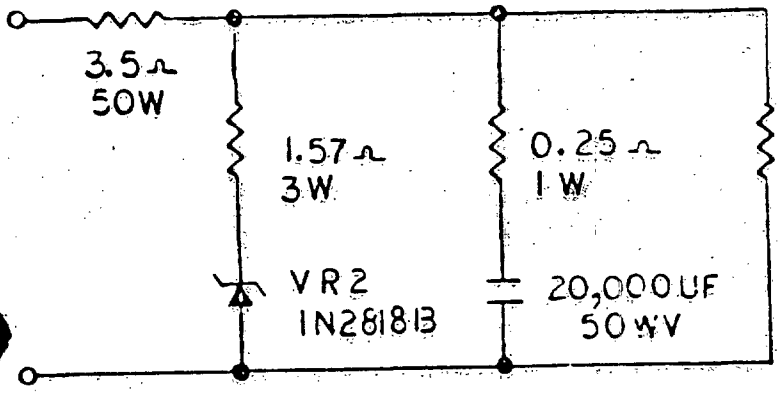
Figure 3. Time Load Study

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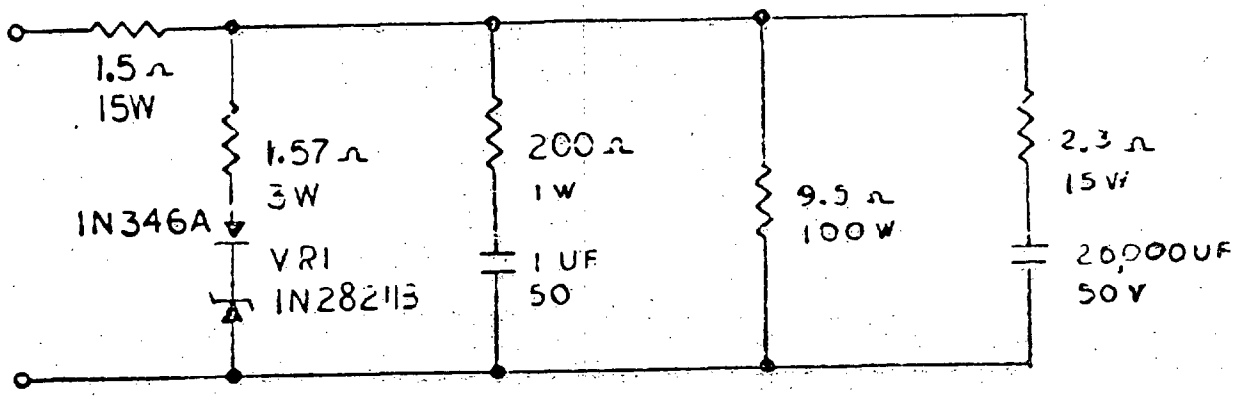
D1, D2 - UNITRODE UES201



TAKE VOLTAGE READINGS BETWEEN A AND B



USE FOR LOADS 1 & 2



USE FOR LOAD 3

Figure 4.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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A. GIVE PARAGRAPH NUMBER AND WORDING.

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